Trigonometric Ratio

Mind map

Basic identities
- $\sin^2 + \cos^2 = 1$, $\sec^2 - \tan^2 = 1$
- $\sin^4 + \cos^4$, $\sin^6 + \cos^6$
- Replace 1 to factorise

Angles and quadrants
- $\pi = 180°$; no symbol = radian
- Signs: All-Sin-Tan-Cos

Reduction formulae
- $n\frac{\pi}{2} \pm \theta$: odd n co-ratio
- Complementary: $\tan A \tan B = 1$
- Sign of $\sin A \pm \cos A$

Compound angles
- $\sin(A \pm B), \cos(A \pm B), \tan(A \pm B)$
- $\sin^2 A - \sin^2 B$ form
- $\tan(\frac{\pi}{4} \pm A)$

$C + D$ formulae
- Product $\leftrightarrow$ sum
- Pair terms, cancel common factor

2A and half angles
- $\sin 2A, \cos 2A, \tan 2A$ in tan
- $1 \pm \cos A$, $1 \pm \sin A$
- $|\sin A \pm \cos A| = \sqrt{1 \pm \sin 2A}$

3A formulae
- $\sin 3A = 3\sin A - 4\sin^3 A$
- $\cos 3A = 4\cos^3 A - 3\cos A$

Standard values
- $15°, 22.5°, 75°$
- $\sin 18° = \frac{\sqrt{5}-1}{4}$, $\cos 36° = \frac{\sqrt{5}+1}{4}$

Six important results
- $\cot A \mp \tan A$
- $\theta, 60° \mp \theta$ products
- $(1 + \tan A)(1 + \tan B) = 2$

Special tools
- Componendo-dividendo
- $\tan(\theta_1 + \cdots)$: $\frac{S_1 - S_3}{1 - S_2}$
- Continued product of cosines

Triangle identities
- $\sum \tan A = \prod \tan A$
- $\sum \sin 2A = 4 \prod \sin A$
- $\sum \sin^2 A = 2$: right angle

Graphs, range type 1
- Basic ranges and periods
- $af + b$, f^2, $\sqrt{f}$, $\frac{1}{f}$

Range types 2, 3
- Quadratic in $\sin x$: complete square
- $a\sin x + b\cos x$: $\pm\sqrt{a^2 + b^2}$
- $|\sin x| + |\cos x| \in [1, \sqrt{2}]$

Range type 4
- $a^2\tan^2 + b^2\cot^2 \geq 2ab$
- $\sin^2$ form: check $\sin^2 x = \frac{b}{a} \leq 1$

Summation of series
- Angles in A.P.: $\frac{\sin(n\beta/2)}{\sin(\beta/2)}$ formula
- Telescoping: $\tan A - \tan B$ split

Inequalities, substitution
- Extreme at $A = B = C$
- $x = r\cos\theta$, $y = r\sin\theta$